CSE145 - Introduction to Computing

Name	Ahmadhussain
ERP No.	33440

R	V	B	n
40	0	101000	7

GitHub repository: https://github.com/AhmadNixor/ITC-Fall-2026---33440

Q1 - Divisible by 3

Task A

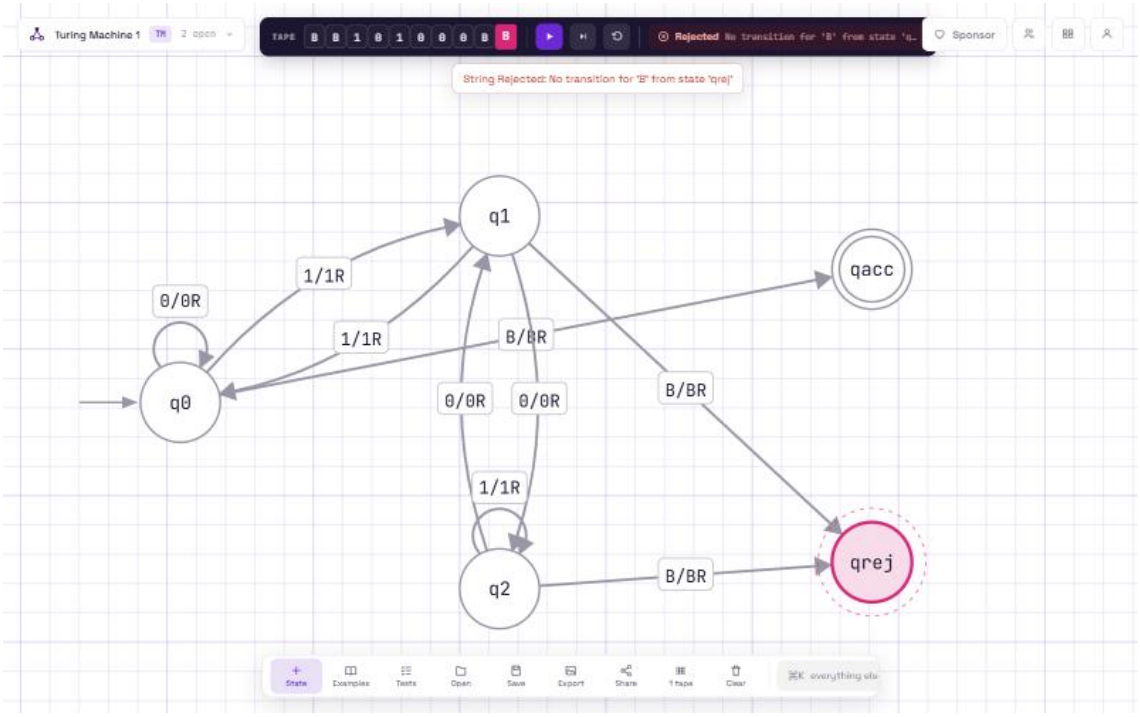
the TM prog reads the binary num from left to right and NEVER changes the tape, all it needs is to remember the Rem of the num so far, when divided by 3, so it has 3 states, q0 for rem 0, q1 for remainder 1 and q2 for rem 2,reading a bit doubles the num and adds the bit, so Rem is (2 x old remainder + bit) mod 3, when it reaches the blank, it accepts if it is in q0 and rejects else, the empty string stays in q0, so it is accepted, which is correct, because 0 is divisible by 3

Task B

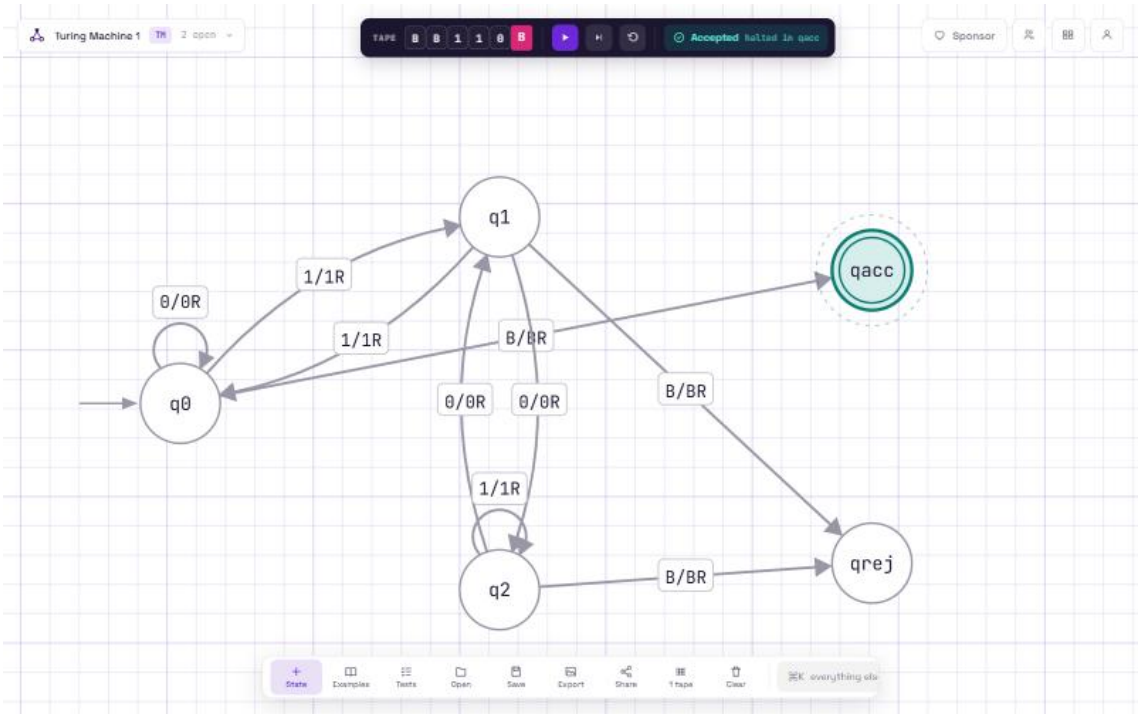
State	Read	Write	Move	Next state
q0	0	0	R	q0
q0	1	1	R	q1
q0	B	B	R	qacc
q1	0	0	R	q2
q1	1	1	R	q0
q1	B	B	R	qrej
q2	0	0	R	q1
q2	1	1	R	q2
q2	B	B	R	qrej

Q1 - Task C

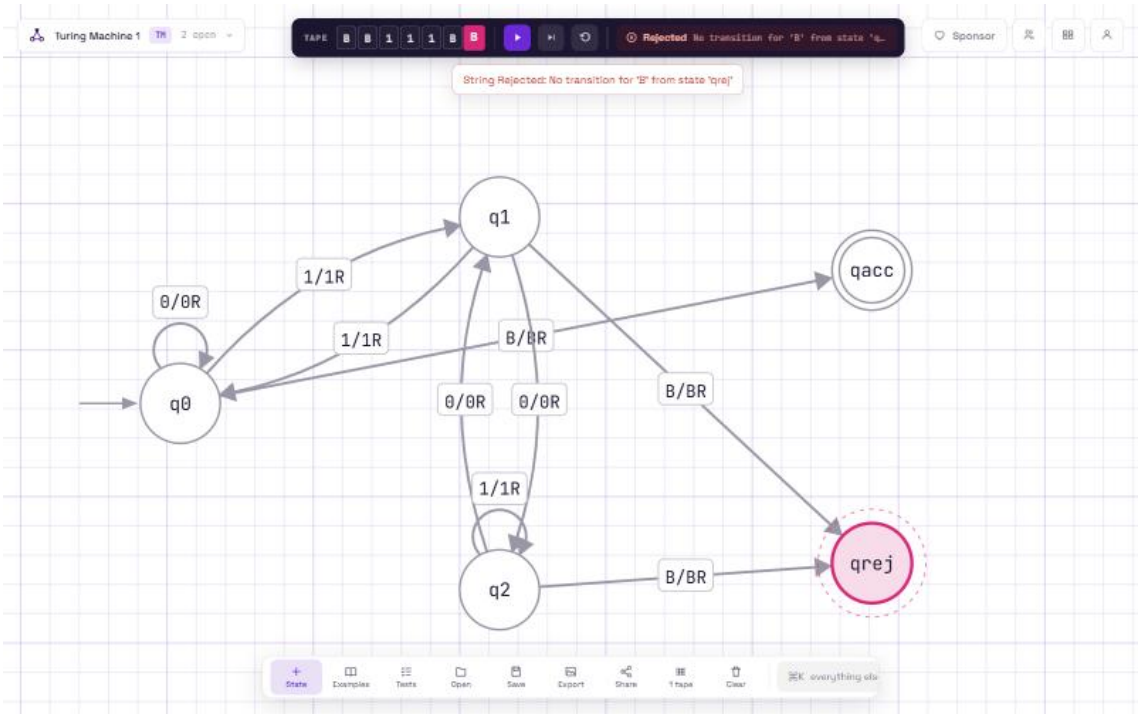
Input	Value	Result	Correct?
101000 (my B)	40	Rejected	Yes, 40 rem 1
110	6	Accepted	Yes, 6 = 3 x 2
111	7	Rejected	Yes, 7 rem 1



101000 - rejected



110 - accepted



111 - rejected

Task D

the TM ONLY works for 3 cuz its 3 states are the 3 possible remainders mod 3, and EVERY arrow was done with mod 3, for a different M, like 5, i would need 5 states (1 per rem) and the arrows would change to (2 x remainder + bit) mod 5, ONLY q0 would still lead to accept

Q2 - a^n b^n

Task A

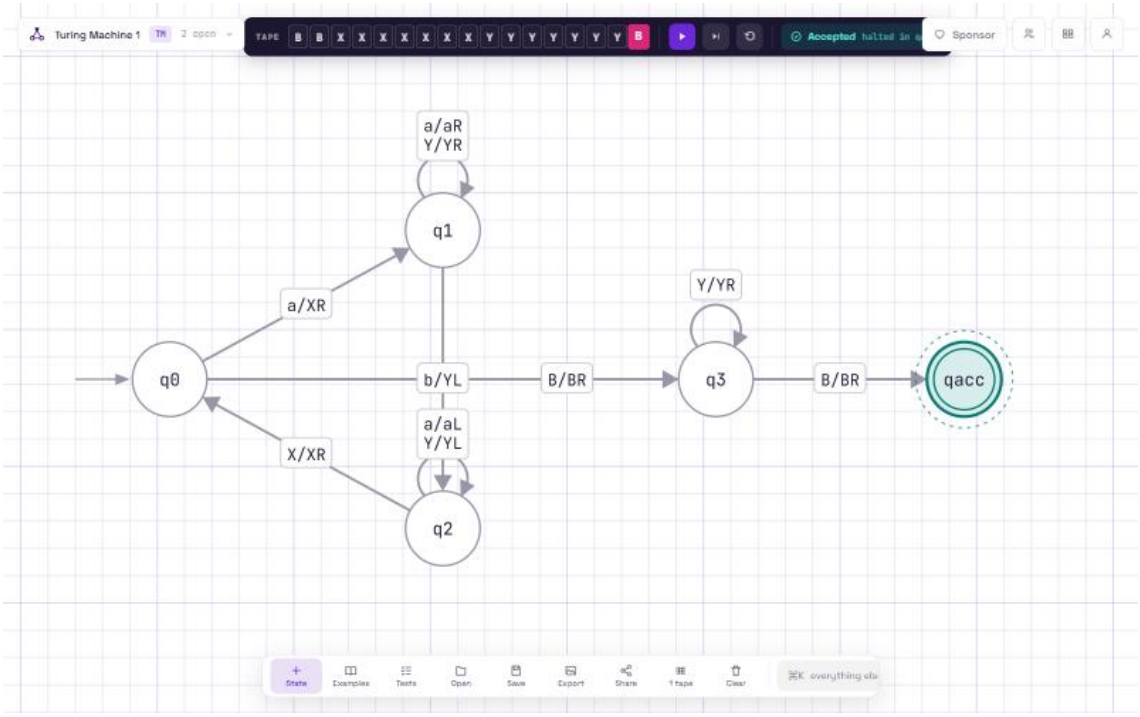
i cross off 1 a and 1 b at a time, in q0 the 1st a is turned to X, then q1 scans right past the other a's and any Y's, until it finds a b, that b is turned to Y, and q2 scans back left, until it hits the X, then steps right, so q0 can start on the nxt a, when q0 sees a Y instead of an a, all the a's are used up, so q3 scans right to check that ONLY Y's are left, if it reaches the blank i accept, a leftover b, or an a with no b to pair with, leaves no matching transition, so the string is rejected

Task B

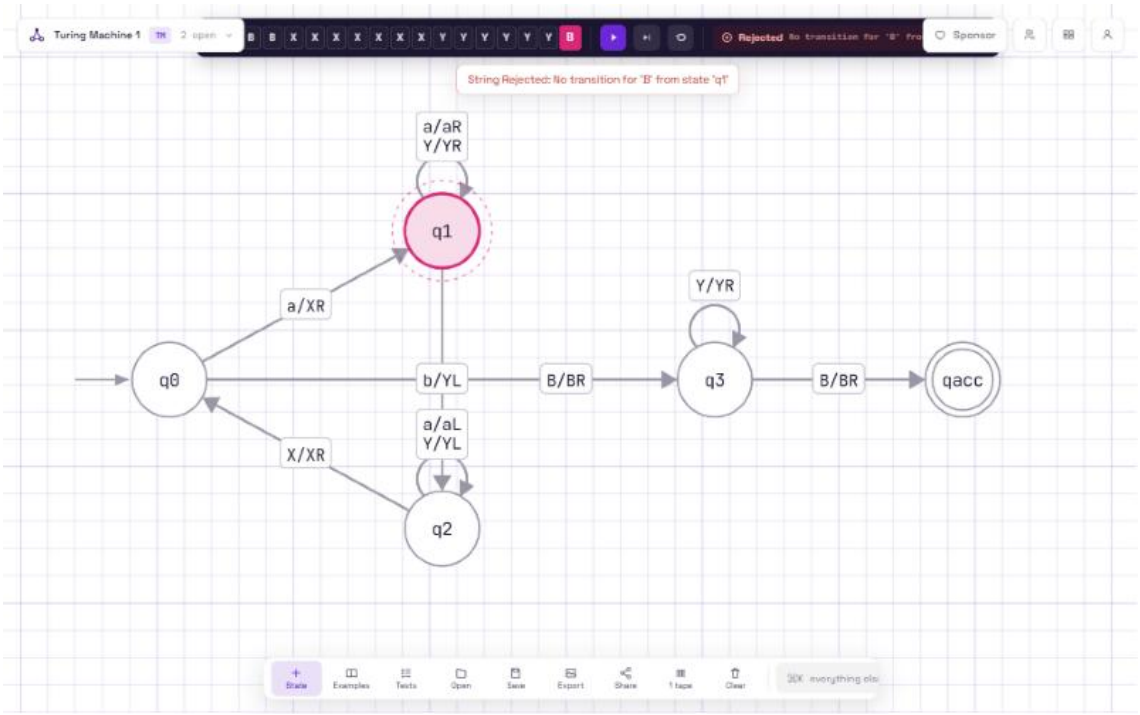
State	Read	Write	Move	Next state
q0	a	X	R	q1
q0	Y	Y	R	q3
q0	B	B	R	qacc
q1	a	a	R	q1
q1	Y	Y	R	q1
q1	b	Y	L	q2
q2	a	a	L	q2
q2	Y	Y	L	q2
q2	X	X	R	q0
q3	Y	Y	R	q3
q3	B	B	R	qacc

Q2 - Task C

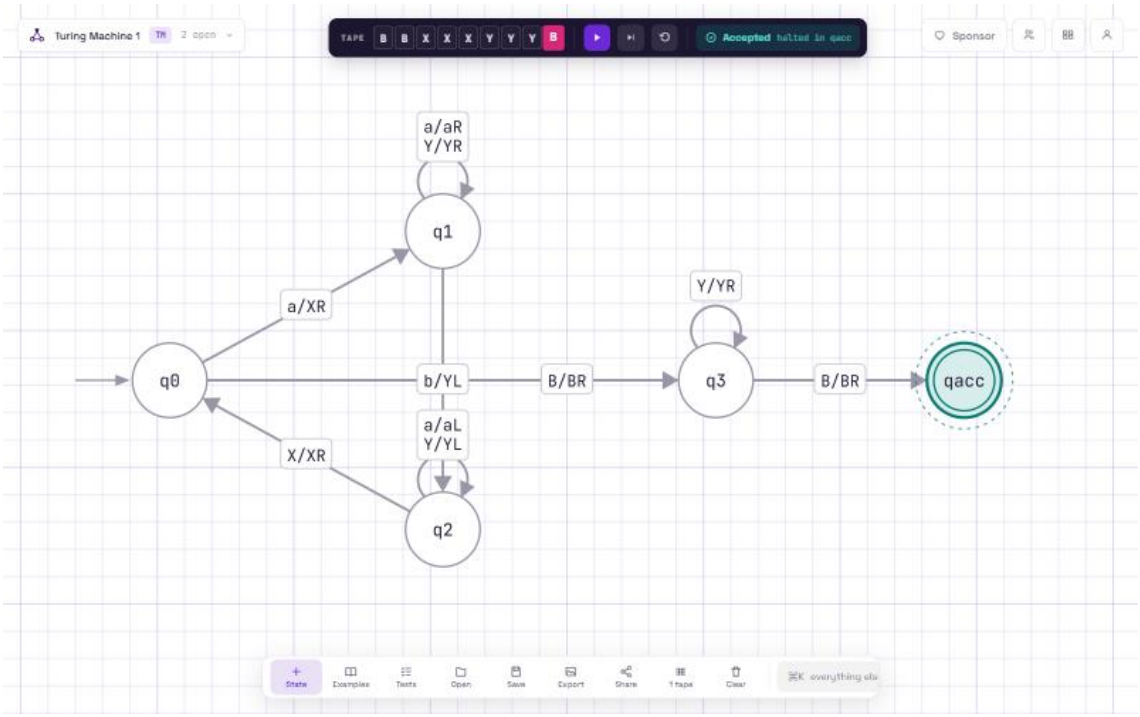
Input	Result	In lang?
aaaaaaabbbbbbb (7 a, 7 b)	Accepted	Yes
aaaaaaabbbbbb (7 a, 6 b)	Rejected	No, 1 b missing
aaabbb	Accepted	Yes, 3 a 3 b



a^7 b^7 - accepted



a^7 b^6 - rejected



aaabbb - accepted

Task D

the most likely 1-off error is right after crossing off a b, q2 scans left, until it sees X, and it must then step right once, so that q0 starts on the nxt a and NOT on the X, the other risky spot is the end, where q3 has to confirm that no b is left over, i checked it by tracing small cases (empty string, ab, aabb) and then running a^7 b^7, which accepts, and a^7 b^6, which rejects, because q1 runs out of b's

Q3 - Add 1 to a binary number

Task A

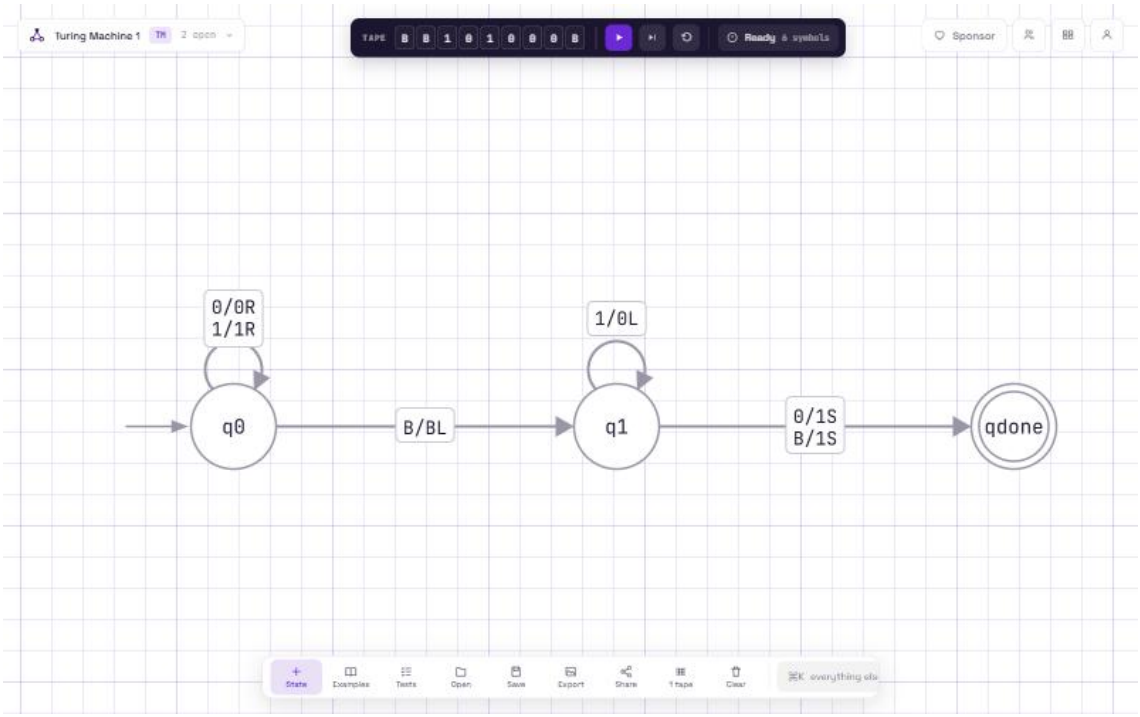
adding 1 works like on paper, so the machine has to start from the right, q0 scans right to the blank, steps back onto the end bit and goes to q1, in q1, a 1 plus the carry becomes 0 and the TM keeps moving left, the 1st 0 it meets becomes 1 and the machine stops, if the carry goes past the left end onto a blank, it writes a 1 there, so 111 becomes 1000

Task B

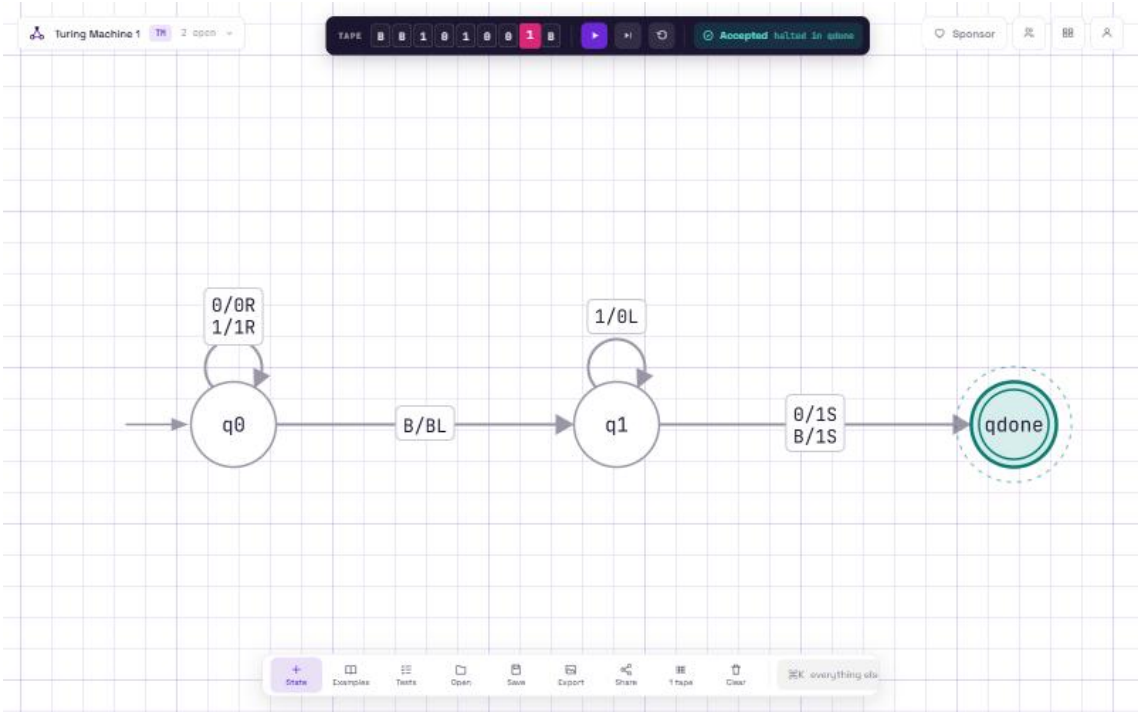
State	Read	Write	Move	Next state
q0	0	0	R	q0
q0	1	1	R	q0
q0	B	B	L	q1
q1	1	0	L	q1
q1	0	1	S	qdone
q1	B	1	S	qdone

Task C

B in binary	101000
B in decimal	32 + 8 = 40
Expected +1	41 = 101001
Final tape	101001 - matches



start, tape is 101000



end, tape is 101001

Task D

a number made of all 1s, like 111, is the edge case, the carry passes through EVERY bit and the answer 1000 is 1 digit longer than the input, so the TM needs the transition that writes a new 1 on the blank at the left, changing ONLY the leftmost bit would fail cuz the carry starts at the right end

Q4 - Palindromes over {a, b}

Task A

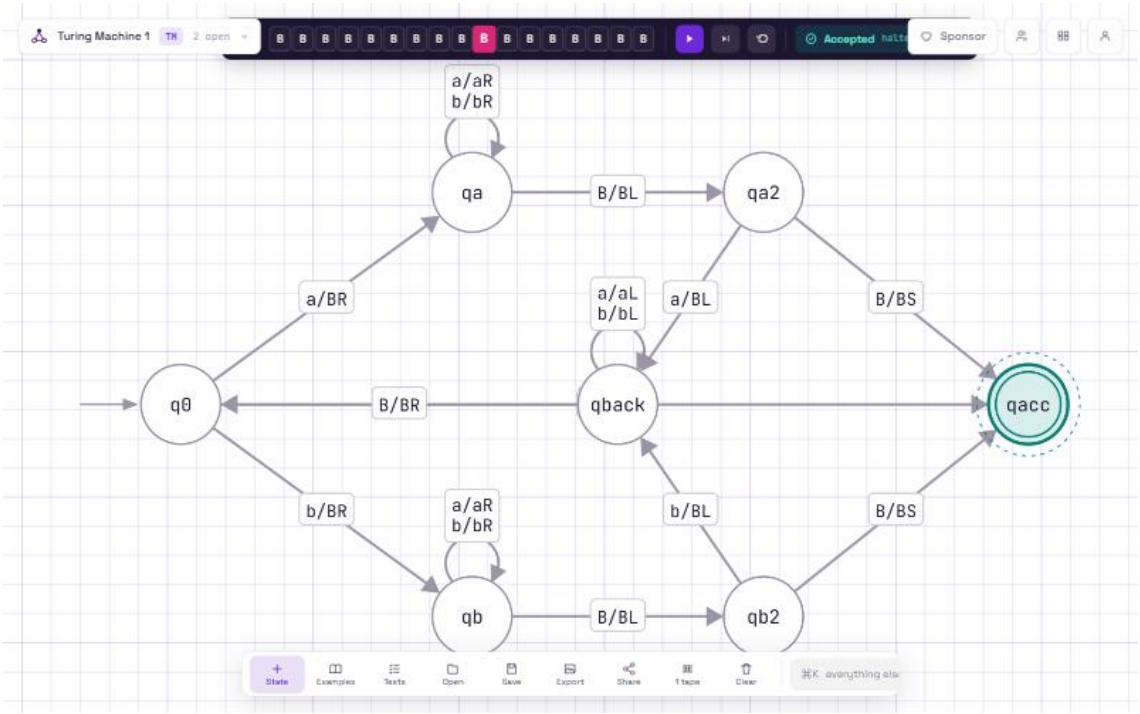
i match the outer letters and work inward, q0 reads the leftmost letter, # it by going to qa or qb, and wipes it, then the head runs to the right end and steps back onto the end letter, if that letter is the same, i wipe it too, move back to the left (qback) and repeat, if it is different there is no transition, so the string is rejected, if everything gets wiped the string is accepted, a single letter left in the middle is matched against the blank, so odd length palindromes are accepted as well

Task B

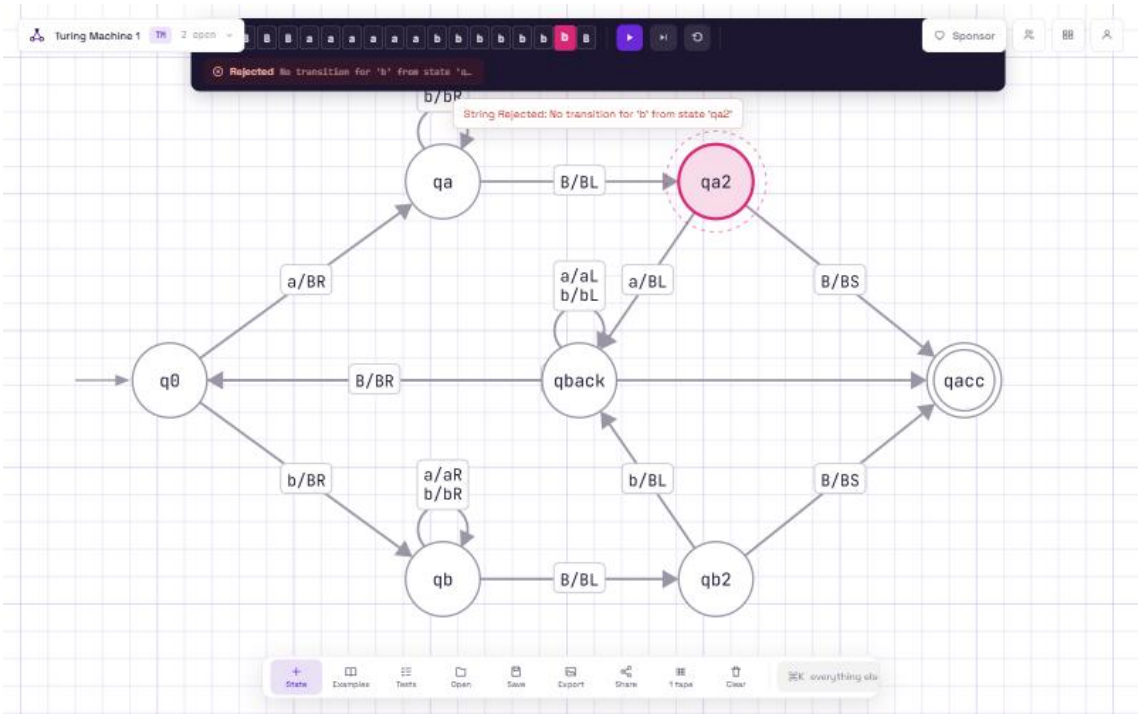
State	Read	Write	Move	Next state
q0	a	B	R	qa
q0	b	B	R	qb
q0	B	B	S	qacc
qa	a	a	R	qa
qa	b	b	R	qa
qa	B	B	L	qa2
qb	a	a	R	qb
qb	b	b	R	qb
qb	B	B	L	qb2
qa2	a	B	L	qback
qa2	B	B	S	qacc
qb2	b	B	L	qback
qb2	B	B	S	qacc
qback	a	a	L	qback
qback	b	b	L	qback
qback	B	B	R	q0

Q4 - Task C

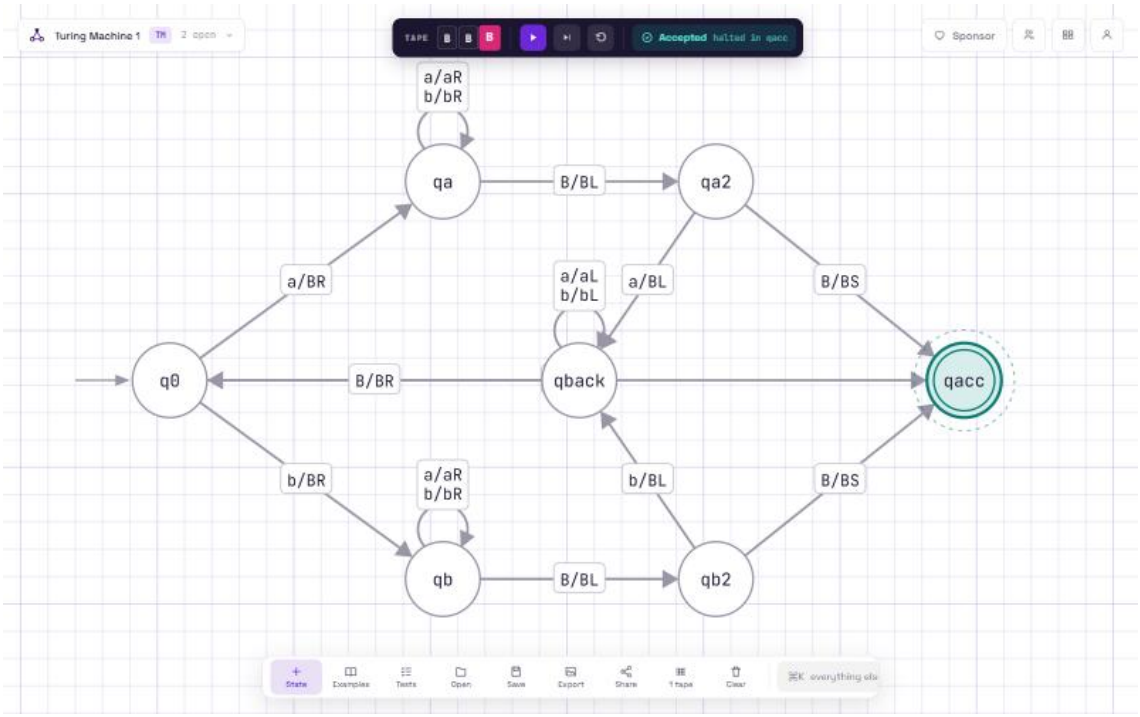
Input	Result	In lang?
abbaabbbbaabba (length 14)	Accepted	Yes, palindrome
aaaaaaabbbbbbb (length 14)	Rejected	No, a front, b back
empty string	Accepted	Yes, empty ok



abbaabbbbaabba - accepted



aaaaaaabbbbbbb - rejected



empty string - accepted

Task D

after the 1st and end letters are matched, the nxt pair to compare is far from the head, the machine can ONLY remember 1 letter at a time, in qa or qb, so it has to go right, come back left and repeat for EVERY pair, with a single pass it could ONLY compare the 1st and end letters, so abba and abaa would look the same to it

Q5 - w#w

Task A

i compare the 2 sides 1 symbol at a time by marking them, q0 marks the 1st unmarked symbol on the left (0 becomes X, 1 becomes Y) and # which it was by going to q0a or q1a, the head then scans right past the # and skips the marked symbols on the right side, the 1st unmarked symbol there has to match, q0b needs a 0 and q1b needs a 1, if it matches, i mark it, go back left (qr, qr2) and repeat, if it does NOT match, or the right side runs out, the string is rejected, when the left side is fully marked, q0 hits the # and qc checks that nothing unmarked is left on the right before accepting

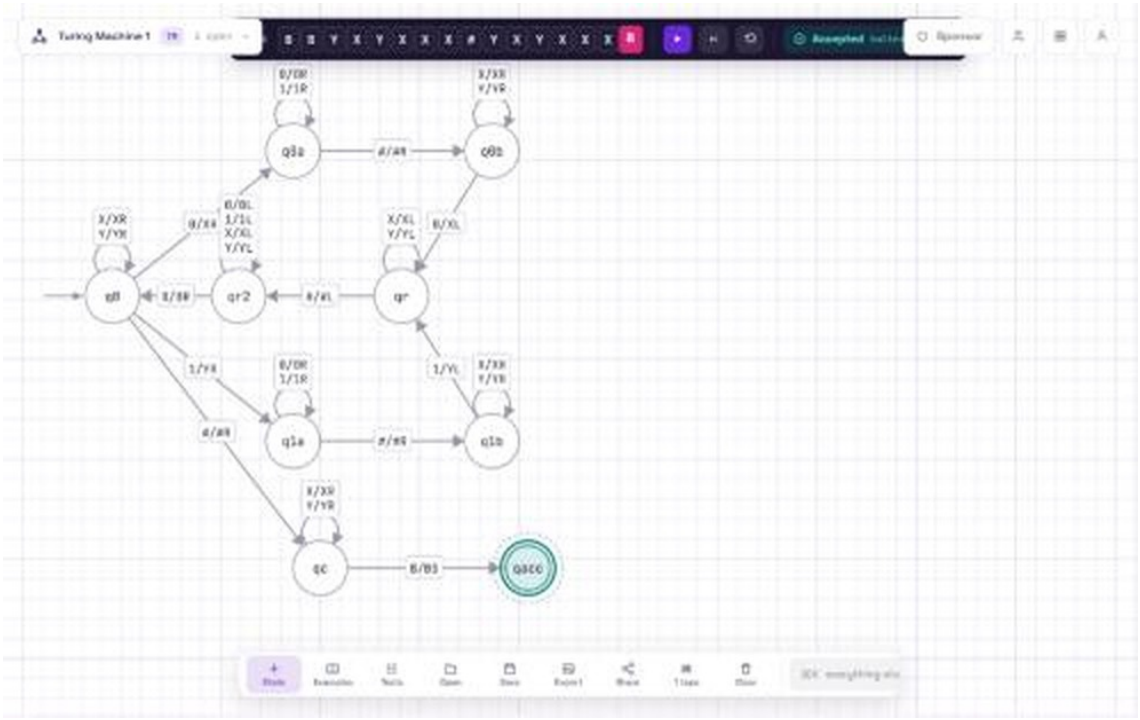
Task B

State	Read	Write	Move	Next state
q0	X	X	R	q0
q0	Y	Y	R	q0
q0	0	X	R	q0a
q0	1	Y	R	q1a
q0	#	#	R	qc
q0a	0	0	R	q0a
q0a	1	1	R	q0a
q0a	#	#	R	q0b
q1a	0	0	R	q1a
q1a	1	1	R	q1a
q1a	#	#	R	q1b
q0b	X	X	R	q0b
q0b	Y	Y	R	q0b
q0b	0	X	L	qr
q1b	X	X	R	q1b
q1b	Y	Y	R	q1b
q1b	1	Y	L	qr
qr	X	X	L	qr
qr	Y	Y	L	qr
qr	#	#	L	qr2
qr2	0	0	L	qr2
qr2	1	1	L	qr2

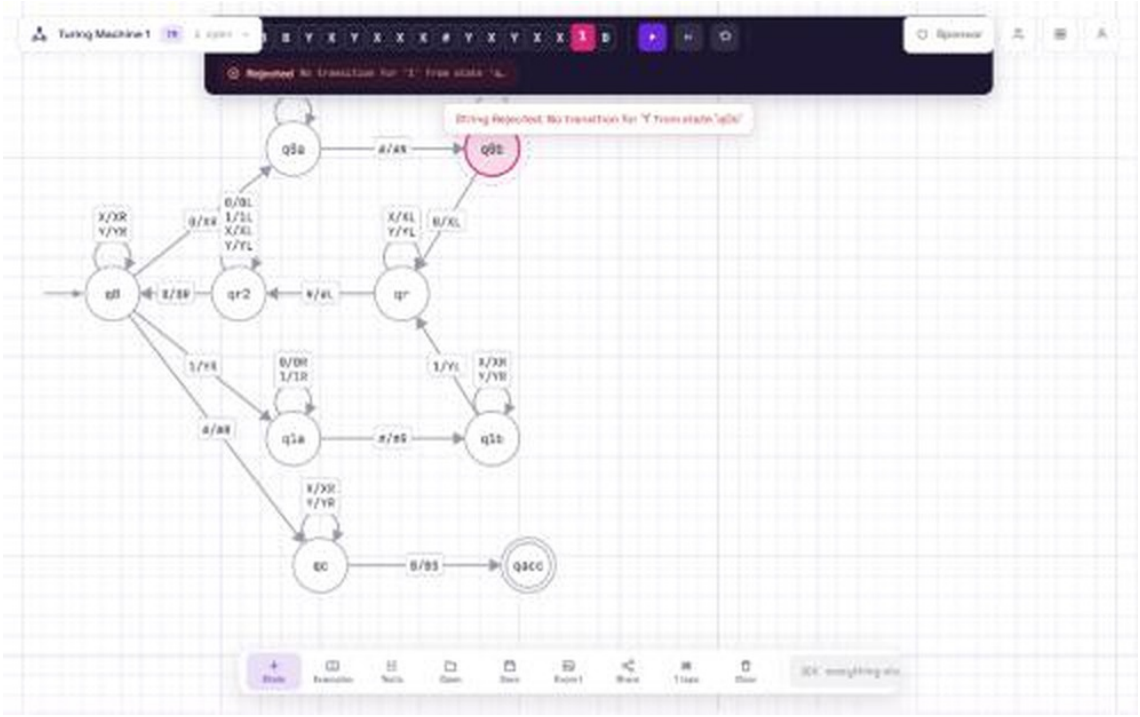
State	Read	Write	Move	Next state
qr2	X	X	L	qr2
qr2	Y	Y	L	qr2
qr2	B	B	R	q0
qc	X	X	R	qc
qc	Y	Y	R	qc
qc	B	B	S	qacc

Q5 - Task C

Input	Result	In lang?
101000#101000	Accepted	Yes, both sides equal
101000#101001	Rejected	No, last symbol differs



101000#101000 - accepted



101000#101001 - rejected

Task D

w can be any length, and a TM with a fixed num of states cant remember a whole string, when the head reaches the #, it would still need all of the left side to compare against, so it remembers 1 symbol at a time (in q0a or q1a), carries it over the #, checks it and goes back for after it, the X and Y marks track what is already matched, which is why the head scans back and forth so much

Q6 - Debug the broken machine

Task A

Step	State	Tape	Head at
0	q_even	_	0
1	q_odd	_ _	1
2	q_reject	_ _ _	2 (halt)

Input "11"

Step	State	Tape	Head at
0	q_even	1 1	0
1	q_odd	1 1	1
2	q_even	1 1 _	2
3	q_odd	1 1 _ _	3
4	q_reject	1 1 _ _ _	4 (halt)

Task B

the wrong row is the 3rd one, q_even, blank, blank, R, q_odd, it should go to q_accept, in both traces the machine hits the blank while in q_even (0 1s and 2 1s, both even), but this row sends it to q_odd, and q_odd rejects on a blank, so the empty string and 11 are rejected, when they should be accepted, the fixed row is (q_even, blank, blank, R, q_accept)

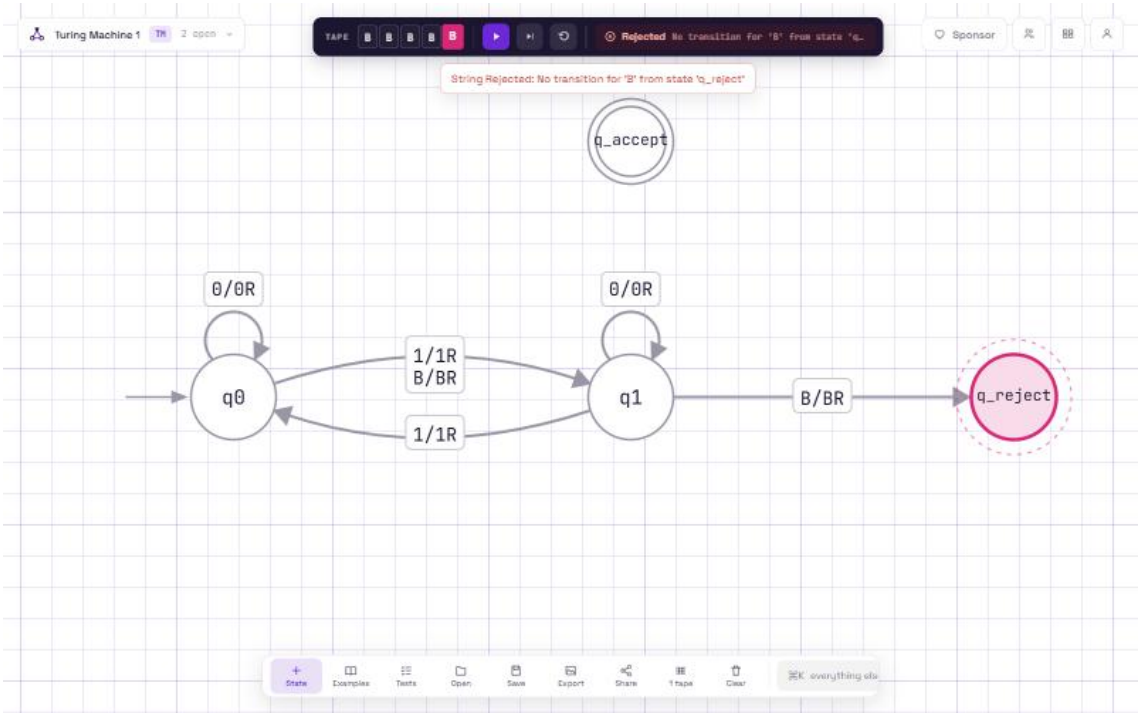
Task D

the strings 1 and 111 would NOT have shown the bug, they should be rejected anyway, and the broken TM rejects everything, so it looks correct on them, ONLY the strings that should be accepted, the empty string and 11, expose it, so i should test inputs that should be accepted as well as ones that should be rejected, and ALWAYS try the empty input

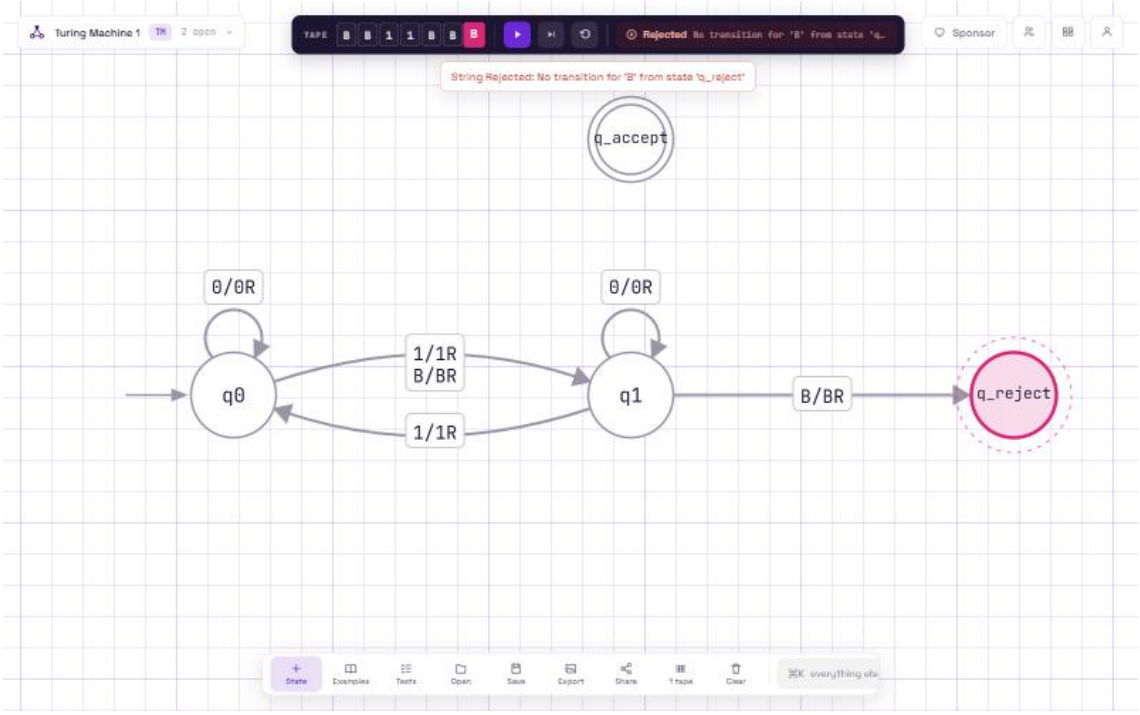
Q6 - Task C

Input	Broken	Fixed	Should be
empty	Reject	Accept	Accept
11	Reject	Accept	Accept
1	Reject	Reject	Reject
111	Reject	Reject	Reject

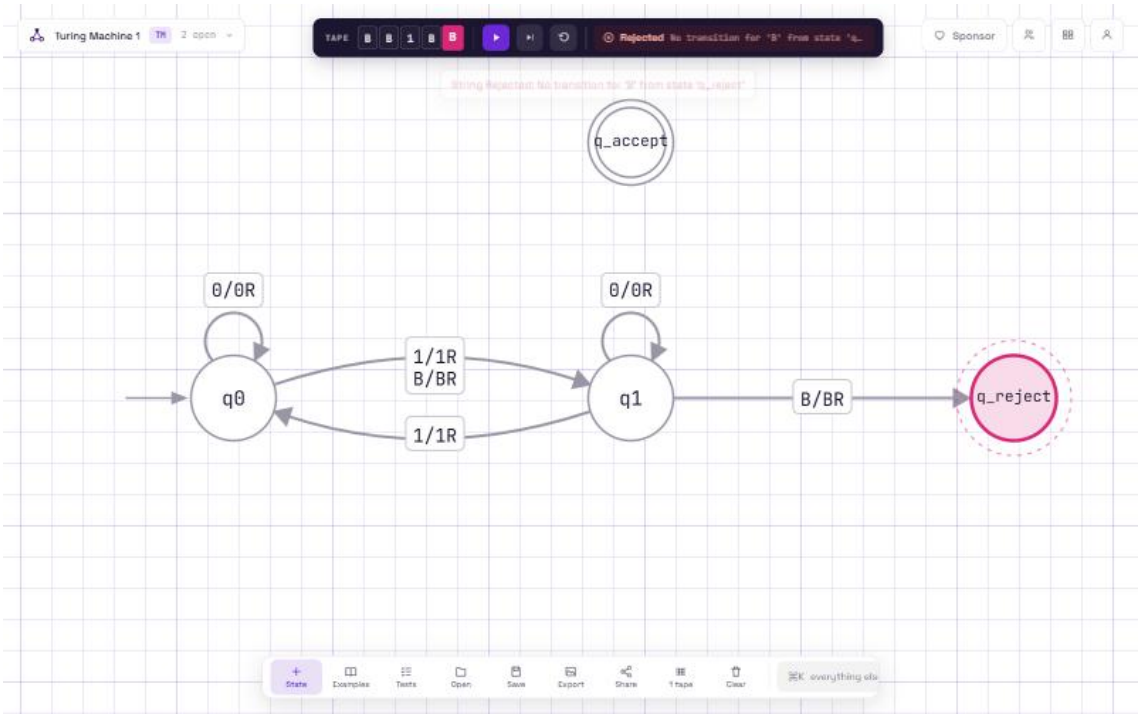
Broken machine



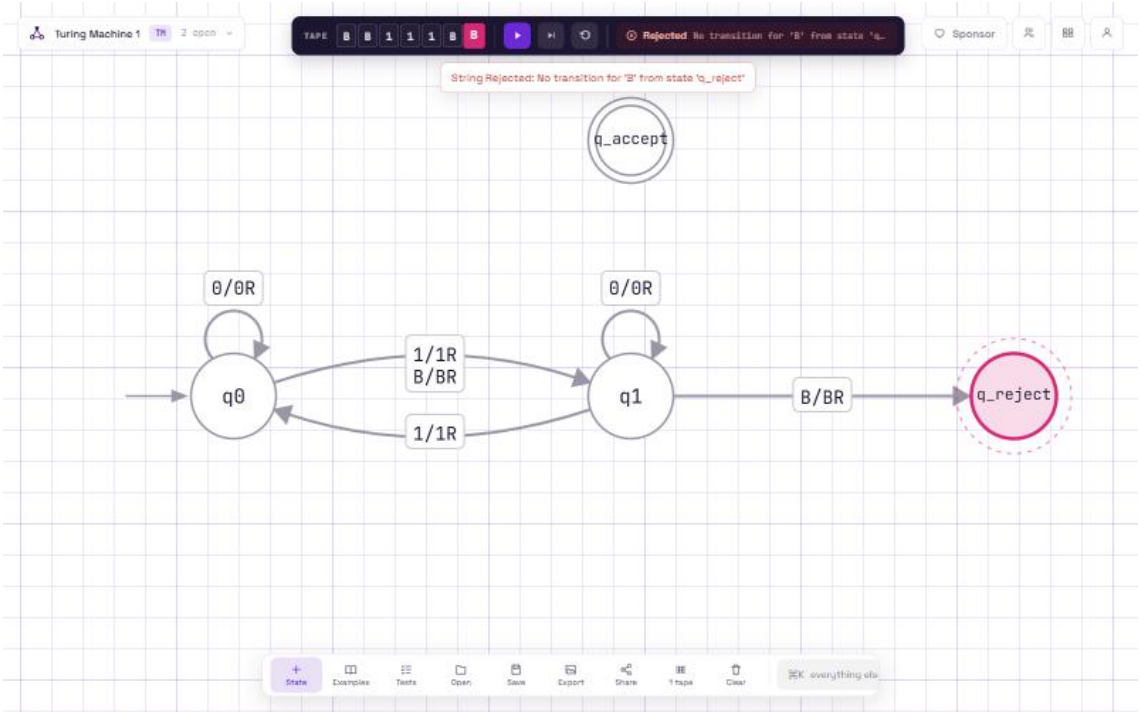
broken, empty - rejected



broken, 11 - rejected

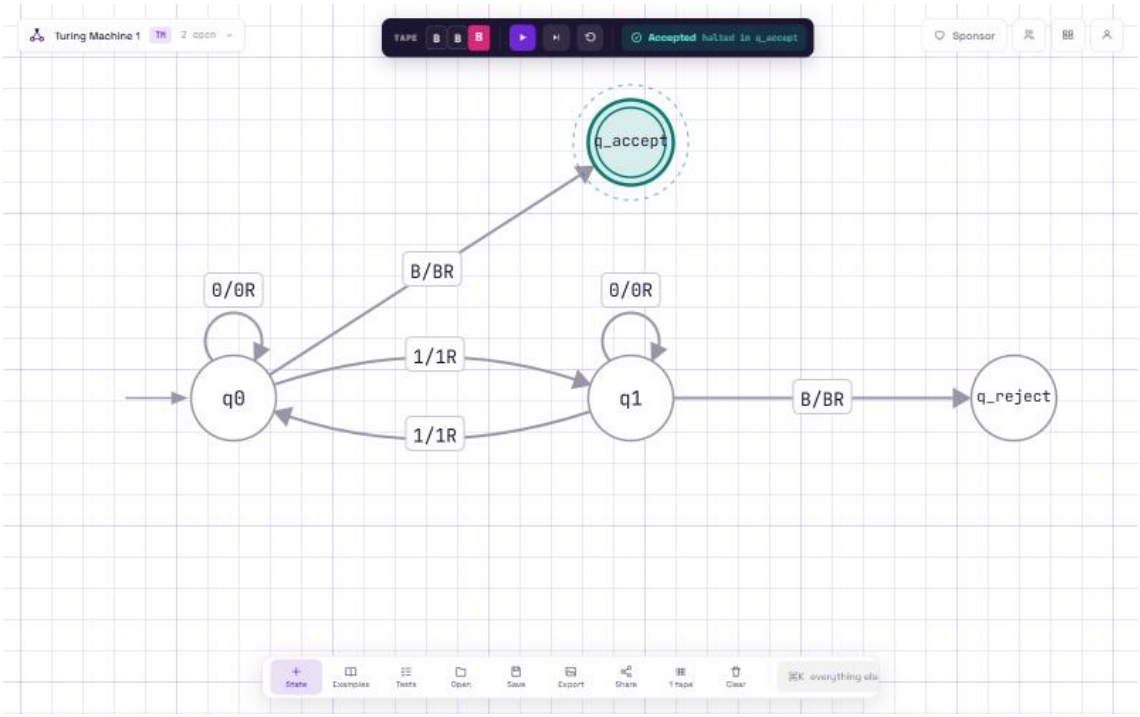


broken, 1 - rejected

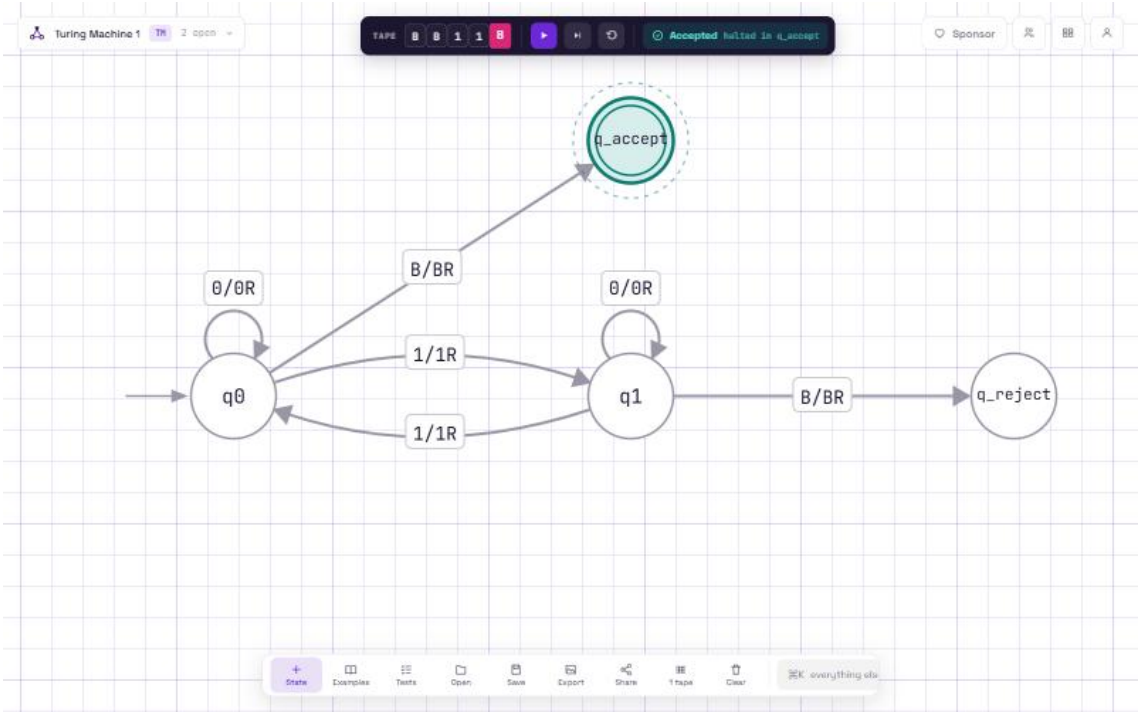


broken, 111 - rejected

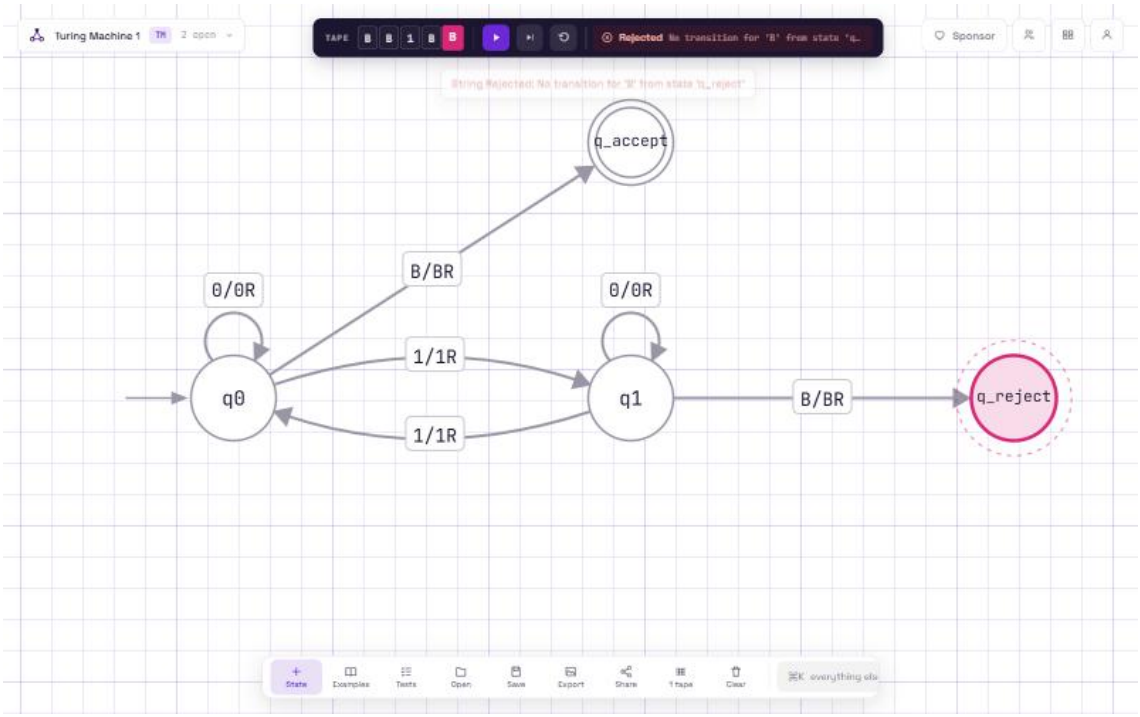
Fixed machine



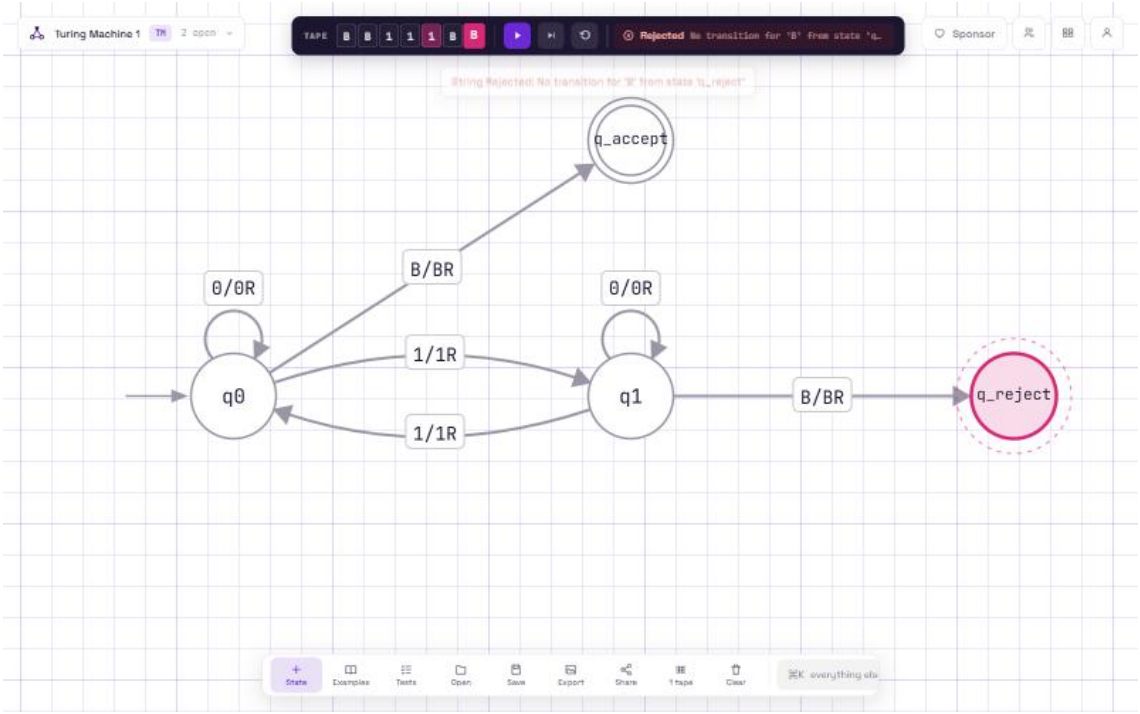
fixed, empty - accepted



fixed, 11 - accepted



fixed, 1 - rejected



fixed, 111 - rejected

JSON files

q1_run1_B.json


```
{"nodeMap":{"q0":{"id":"q0","x":120,"y":260},"q1":{"id":"q1","x":360,"y":120},"q2":{"id":"q2","x":360,"y":400},"qacc":{"
id":"qacc","x":640,"y":160},"qrej":{"id":"qrej","x":640,"y":380}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0"
,"targetid":"q0"},{"label":"1/1R","sourceid":"q0","targetid":"q1"},{"label":"B/BR","sourceid":"q0","targetid":"qacc"}],"
q1":[{"label":"0/0R","sourceid":"q1","targetid":"q2"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","
sourceid":"q1","targetid":"qrej"}],"q2":[{"label":"0/0R","sourceid":"q2","targetid":"q1"},{"label":"1/1R","sourceid":"q2
","targetid":"q2"},{"label":"B/BR","sourceid":"q2","targetid":"qrej"}]},"finalNodes":["qacc"],"nodeNum":5,"automataType"
:"TM","inputString":"101000","stageProps":{"x":20,"y":60,"scale":1.5,"draggable":true},"stateOutput":{},"tapeMode":"sing
le","inputString2":""}
```


q1_run2_div.json


```
{"nodeMap":{"q0":{"id":"q0","x":120,"y":260},"q1":{"id":"q1","x":360,"y":120},"q2":{"id":"q2","x":360,"y":400},"qacc":{"
id":"qacc","x":640,"y":160},"qrej":{"id":"qrej","x":640,"y":380}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0"
,"targetid":"q0"},{"label":"1/1R","sourceid":"q0","targetid":"q1"},{"label":"B/BR","sourceid":"q0","targetid":"qacc"}],"
q1":[{"label":"0/0R","sourceid":"q1","targetid":"q2"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","
sourceid":"q1","targetid":"qrej"}],"q2":[{"label":"0/0R","sourceid":"q2","targetid":"q1"},{"label":"1/1R","sourceid":"q2
","targetid":"q2"},{"label":"B/BR","sourceid":"q2","targetid":"qrej"}]},"finalNodes":["qacc"],"nodeNum":5,"automataType"
:"TM","inputString":"110","stageProps":{"x":20,"y":60,"scale":1.5,"draggable":true},"stateOutput":{},"tapeMode":"single"
,"inputString2":""}
```


q1_run3_notdiv.json


```
{"nodeMap":{"q0":{"id":"q0","x":120,"y":260},"q1":{"id":"q1","x":360,"y":120},"q2":{"id":"q2","x":360,"y":400},"qacc":{"
id":"qacc","x":640,"y":160},"qrej":{"id":"qrej","x":640,"y":380}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0"
,"targetid":"q0"},{"label":"1/1R","sourceid":"q0","targetid":"q1"},{"label":"B/BR","sourceid":"q0","targetid":"qacc"}],"
q1":[{"label":"0/0R","sourceid":"q1","targetid":"q2"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","
sourceid":"q1","targetid":"qrej"}],"q2":[{"label":"0/0R","sourceid":"q2","targetid":"q1"},{"label":"1/1R","sourceid":"q2
","targetid":"q2"},{"label":"B/BR","sourceid":"q2","targetid":"qrej"}]},"finalNodes":["qacc"],"nodeNum":5,"automataType"
:"TM","inputString":"111","stageProps":{"x":20,"y":60,"scale":1.5,"draggable":true},"stateOutput":{},"tapeMode":"single"
,"inputString2":""}
```


q2_run1_n7.json


```
{"nodeMap":{"q0":{"id":"q0","x":120,"y":260},"q1":{"id":"q1","x":340,"y":140},"q2":{"id":"q2","x":340,"y":380},"q3":{"id
":"q3","x":560,"y":260},"qacc":{"id":"qacc","x":760,"y":260}},"transitionMap":{"q0":[{"label":"a/XR","sourceid":"q0","ta
rgetid":"q1"},{"label":"Y/YR","sourceid":"q0","targetid":"q3"},{"label":"B/BR","sourceid":"q0","targetid":"qacc"}],"q1":
[{"label":"a/aR,Y/YR","sourceid":"q1","targetid":"q1"},{"label":"b/YL","sourceid":"q1","targetid":"q2"}],"q2":[{"label":
"a/aL,Y/YL","sourceid":"q2","targetid":"q2"},{"label":"X/XR","sourceid":"q2","targetid":"q0"}],"q3":[{"label":"Y/YR","so
urceid":"q3","targetid":"q3"},{"label":"B/BR","sourceid":"q3","targetid":"qacc"}]},"finalNodes":["qacc"],"nodeNum":5,"au
tomataType":"TM","inputString":"aaaaaaabbbbbbb","stageProps":{"x":20,"y":60,"scale":1.4,"draggable":true},"stateOutput":
{},"tapeMode":"single","inputString2":""}
```


q2_run2_short.json


```
{"nodeMap":{"q0":{"id":"q0","x":120,"y":260},"q1":{"id":"q1","x":340,"y":140},"q2":{"id":"q2","x":340,"y":380},"q3":{"id
":"q3","x":560,"y":260},"qacc":{"id":"qacc","x":760,"y":260}},"transitionMap":{"q0":[{"label":"a/XR","sourceid":"q0","ta
rgetid":"q1"},{"label":"Y/YR","sourceid":"q0","targetid":"q3"},{"label":"B/BR","sourceid":"q0","targetid":"qacc"}],"q1":
[{"label":"a/aR,Y/YR","sourceid":"q1","targetid":"q1"},{"label":"b/YL","sourceid":"q1","targetid":"q2"}],"q2":[{"label":
"a/aL,Y/YL","sourceid":"q2","targetid":"q2"},{"label":"X/XR","sourceid":"q2","targetid":"q0"}],"q3":[{"label":"Y/YR","so
urceid":"q3","targetid":"q3"},{"label":"B/BR","sourceid":"q3","targetid":"qacc"}]},"finalNodes":["qacc"],"nodeNum":5,"au
tomataType":"TM","inputString":"aaaaaaabbbbbb","stageProps":{"x":20,"y":60,"scale":1.4,"draggable":true},"stateOutput":{
},"tapeMode":"single","inputString2":""}
```


q2_run3_extra.json


```
{"nodeMap":{"q0":{"id":"q0","x":120,"y":260},"q1":{"id":"q1","x":340,"y":140},"q2":{"id":"q2","x":340,"y":380},"q3":{"id
":"q3","x":560,"y":260},"qacc":{"id":"qacc","x":760,"y":260}},"transitionMap":{"q0":[{"label":"a/XR","sourceid":"q0","ta
rgetid":"q1"},{"label":"Y/YR","sourceid":"q0","targetid":"q3"},{"label":"B/BR","sourceid":"q0","targetid":"qacc"}],"q1":
[{"label":"a/aR,Y/YR","sourceid":"q1","targetid":"q1"},{"label":"b/YL","sourceid":"q1","targetid":"q2"}],"q2":[{"label":
"a/aL,Y/YL","sourceid":"q2","targetid":"q2"},{"label":"X/XR","sourceid":"q2","targetid":"q0"}],"q3":[{"label":"Y/YR","so
urceid":"q3","targetid":"q3"},{"label":"B/BR","sourceid":"q3","targetid":"qacc"}]},"finalNodes":["qacc"],"nodeNum":5,"au
tomataType":"TM","inputString":"aaabbb","stageProps":{"x":20,"y":60,"scale":1.4,"draggable":true},"stateOutput":{},"tape
Mode":"single","inputString2":""}
```


q3_run1_B.json


```
{"nodeMap":{"q0":{"id":"q0","x":150,"y":260},"q1":{"id":"q1","x":420,"y":260},"qdone":{"id":"qdone","x":700,"y":260}},"t
ransitionMap":{"q0":[{"label":"0/0R,1/1R","sourceid":"q0","targetid":"q0"},{"label":"B/BL","sourceid":"q0","targetid":"q
1"}],"q1":[{"label":"1/0L","sourceid":"q1","targetid":"q1"},{"label":"0/1S,B/1S","sourceid":"q1","targetid":"qdone"}]},"
finalNodes":["qdone"],"nodeNum":3,"automataType":"TM","inputString":"101000","stageProps":{"x":20,"y":60,"scale":1.6,"dr
aggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}
```


q4_run1_member.json


```
{"nodeMap":{"q0":{"id":"q0","x":120,"y":260},"qa":{"id":"qa","x":340,"y":100},"qb":{"id":"qb","x":340,"y":420},"qa2":{"i
d":"qa2","x":560,"y":100},"qb2":{"id":"qb2","x":560,"y":420},"qback":{"id":"qback","x":450,"y":260},"qacc":{"id":"qacc",
"x":760,"y":260}},"transitionMap":{"q0":[{"label":"a/BR","sourceid":"q0","targetid":"qa"},{"label":"b/BR","sourceid":"q0
","targetid":"qb"},{"label":"B/BS","sourceid":"q0","targetid":"qacc"}],"qa":[{"label":"a/aR,b/bR","sourceid":"qa","targe
tid":"qa"},{"label":"B/BL","sourceid":"qa","targetid":"qa2"}],"qb":[{"label":"a/aR,b/bR","sourceid":"qb","targetid":"qb"
},{"label":"B/BL","sourceid":"qb","targetid":"qb2"}],"qa2":[{"label":"a/BL","sourceid":"qa2","targetid":"qback"},{"label
":"B/BS","sourceid":"qa2","targetid":"qacc"}],"qb2":[{"label":"b/BL","sourceid":"qb2","targetid":"qback"},{"label":"B/BS
","sourceid":"qb2","targetid":"qacc"}],"qback":[{"label":"a/aL,b/bL","sourceid":"qback","targetid":"qback"},{"label":"B/
BR","sourceid":"qback","targetid":"q0"}]},"finalNodes":["qacc"],"nodeNum":7,"automataType":"TM","inputString":"abbaabbbb
aabba","stageProps":{"x":20,"y":60,"scale":1.4,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}
```


q4_run2_nonmember.json

{"nodeMap":{"q0":{"id":"q0","x":120,"y":260},"qa":{"id":"qa","x":340,"y":100},"qb":{"id":"qb","x":340,"y":420},"qa2":{"id":"qa2","x":560,"y":100},"qb2":{"id":"qb2","x":560,"y":420},"qback":{"id":"qback","x":450,"y":260},"qacc":{"id":"qacc","x":760,"y":260}},"transitionMap":{"q0":[{"label":"a/BR","sourceid":"q0","targetid":"qa"},{"label":"b/BR","sourceid":"q0","targetid":"qb"},{"label":"B/BS","sourceid":"q0","targetid":"qacc"}],"qa":[{"label":"a/aR,b/bR","sourceid":"qa","targetid":"qa"},{"label":"B/BL","sourceid":"qa","targetid":"qa2"}],"qb":[{"label":"a/aR,b/bR","sourceid":"qb","targetid":"qb"},{"label":"B/BL","sourceid":"qb","targetid":"qb2"}],"qa2":[{"label":"a/BL","sourceid":"qa2","targetid":"qback"},{"label":"B/BS","sourceid":"qa2","targetid":"qacc"}],"qb2":[{"label":"b/BL","sourceid":"qb2","targetid":"qback"},{"label":"B/BS","sourceid":"qb2","targetid":"qacc"}],"qback":[{"label":"a/aL,b/bL","sourceid":"qback","targetid":"qback"},{"label":"B/BR","sourceid":"qback","targetid":"q0"}]},"finalNodes":["qacc"],"nodeNum":7,"automataType":"TM","inputString":"aaaaaaabbbbbb","stageProps":{"x":20,"y":60,"scale":1.4,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q4_run3_empty.json

{"nodeMap":{"q0":{"id":"q0","x":120,"y":260},"qa":{"id":"qa","x":340,"y":100},"qb":{"id":"qb","x":340,"y":420},"qa2":{"id":"qa2","x":560,"y":100},"qb2":{"id":"qb2","x":560,"y":420},"qback":{"id":"qback","x":450,"y":260},"qacc":{"id":"qacc","x":760,"y":260}},"transitionMap":{"q0":[{"label":"a/BR","sourceid":"q0","targetid":"qa"},{"label":"b/BR","sourceid":"q0","targetid":"qb"},{"label":"B/BS","sourceid":"q0","targetid":"qacc"}],"qa":[{"label":"a/aR,b/bR","sourceid":"qa","targetid":"qa"},{"label":"B/BL","sourceid":"qa","targetid":"qa2"}],"qb":[{"label":"a/aR,b/bR","sourceid":"qb","targetid":"qb"},{"label":"B/BL","sourceid":"qb","targetid":"qb2"}],"qa2":[{"label":"a/BL","sourceid":"qa2","targetid":"qback"},{"label":"B/BS","sourceid":"qa2","targetid":"qacc"}],"qb2":[{"label":"b/BL","sourceid":"qb2","targetid":"qback"},{"label":"B/BS","sourceid":"qb2","targetid":"qacc"}],"qback":[{"label":"a/aL,b/bL","sourceid":"qback","targetid":"qback"},{"label":"B/BR","sourceid":"qback","targetid":"q0"}]},"finalNodes":["qacc"],"nodeNum":7,"automataType":"TM","inputString":"","stageProps":{"x":20,"y":60,"scale":1.4,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q5_run1_member.json

{"nodeMap":{"q0":{"id":"q0","x":100,"y":260},"q0a":{"id":"q0a","x":300,"y":100},"q1a":{"id":"q1a","x":300,"y":420},"q0b":{"id":"q0b","x":520,"y":100},"q1b":{"id":"q1b","x":520,"y":420},"qr":{"id":"qr","x":420,"y":260},"qr2":{"id":"qr2","x":240,"y":260},"qc":{"id":"qc","x":330,"y":560},"qacc":{"id":"qacc","x":560,"y":560}},"transitionMap":{"q0":[{"label":"X/XR,Y/YR","sourceid":"q0","targetid":"q0"},{"label":"0/XR","sourceid":"q0","targetid":"q0a"},{"label":"1/YR","sourceid":"q0","targetid":"q1a"},{"label":"#/#R","sourceid":"q0","targetid":"qc"}],"q0a":[{"label":"0/0R,1/1R","sourceid":"q0a","targetid":"q0a"},{"label":"#/#R","sourceid":"q0a","targetid":"q0b"}],"q1a":[{"label":"0/0R,1/1R","sourceid":"q1a","targetid":"q1a"},{"label":"#/#R","sourceid":"q1a","targetid":"q1b"}],"q0b":[{"label":"X/XR,Y/YR","sourceid":"q0b","targetid":"q0b"},{"label":"0/XL","sourceid":"q0b","targetid":"qr"}],"q1b":[{"label":"X/XR,Y/YR","sourceid":"q1b","targetid":"q1b"},{"label":"1/YL","sourceid":"q1b","targetid":"qr"}],"qr":[{"label":"X/XL,Y/YL","sourceid":"qr","targetid":"qr"},{"label":"#/#L","sourceid":"qr","targetid":"qr2"}],"qr2":[{"label":"0/0L,1/1L,X/XL,Y/YL","sourceid":"qr2","targetid":"qr2"},{"label":"B/BR","sourceid":"qr2","targetid":"q0"}],"qc":[{"label":"X/XR,Y/YR","sourceid":"qc","targetid":"qc"},{"label":"B/BS","sourceid":"qc","targetid":"qacc"}]},"finalNodes":["qacc"],"nodeNum":9,"automataType":"TM","inputString":"101000#101000","stageProps":{"x":20,"y":60,"scale":0.95,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q5_run2_nonmember.json

{"nodeMap":{"q0":{"id":"q0","x":100,"y":260},"q0a":{"id":"q0a","x":300,"y":100},"q1a":{"id":"q1a","x":300,"y":420},"q0b":{"id":"q0b","x":520,"y":100},"q1b":{"id":"q1b","x":520,"y":420},"qr":{"id":"qr","x":420,"y":260},"qr2":{"id":"qr2","x":240,"y":260},"qc":{"id":"qc","x":330,"y":560},"qacc":{"id":"qacc","x":560,"y":560}},"transitionMap":{"q0":[{"label":"X/XR,Y/YR","sourceid":"q0","targetid":"q0"},{"label":"0/XR","sourceid":"q0","targetid":"q0a"},{"label":"1/YR","sourceid":"q0","targetid":"q1a"},{"label":"#/#R","sourceid":"q0","targetid":"qc"}],"q0a":[{"label":"0/0R,1/1R","sourceid":"q0a","targetid":"q0a"},{"label":"#/#R","sourceid":"q0a","targetid":"q0b"}],"q1a":[{"label":"0/0R,1/1R","sourceid":"q1a","targetid":"q1a"},{"label":"#/#R","sourceid":"q1a","targetid":"q1b"}],"q0b":[{"label":"X/XR,Y/YR","sourceid":"q0b","targetid":"q0b"},{"label":"0/XL","sourceid":"q0b","targetid":"qr"}],"q1b":[{"label":"X/XR,Y/YR","sourceid":"q1b","targetid":"q1b"},{"label":"1/YL","sourceid":"q1b","targetid":"qr"}],"qr":[{"label":"X/XL,Y/YL","sourceid":"qr","targetid":"qr"},{"label":"#/#L","sourceid":"qr","targetid":"qr2"}],"qr2":[{"label":"0/0L,1/1L,X/XL,Y/YL","sourceid":"qr2","targetid":"qr2"},{"label":"B/BR","sourceid":"qr2","targetid":"q0"}],"qc":[{"label":"X/XR,Y/YR","sourceid":"qc","targetid":"qc"},{"label":"B/BS","sourceid":"qc","targetid":"qacc"}]},"finalNodes":["qacc"],"nodeNum":9,"automataType":"TM","inputString":"101000#101001","stageProps":{"x":20,"y":60,"scale":0.95,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q6_broken_empty.json

{"nodeMap":{"q0":{"id":"q0","x":150,"y":240},"q1":{"id":"q1","x":430,"y":240},"q_accept":{"id":"q_accept","x":430,"y":60},"q_reject":{"id":"q_reject","x":700,"y":240}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0","targetid":"q0"},{"label":"1/1R,B/BR","sourceid":"q0","targetid":"q1"}],"q1":[{"label":"0/0R","sourceid":"q1","targetid":"q1"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","sourceid":"q1","targetid":"q_reject"}]},"finalNodes":["q_accept"],"nodeNum":4,"automataType":"TM","inputString":"","stageProps":{"x":20,"y":60,"scale":1.6,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q6_broken_11.json

{"nodeMap":{"q0":{"id":"q0","x":150,"y":240},"q1":{"id":"q1","x":430,"y":240},"q_accept":{"id":"q_accept","x":430,"y":60},"q_reject":{"id":"q_reject","x":700,"y":240}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0","targetid":"q0"},{"label":"1/1R,B/BR","sourceid":"q0","targetid":"q1"}],"q1":[{"label":"0/0R","sourceid":"q1","targetid":"q1"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","sourceid":"q1","targetid":"q_reject"}]},"finalNodes":["q_accept"],"nodeNum":4,"automataType":"TM","inputString":"11","stageProps":{"x":20,"y":60,"scale":1.6,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q6_broken_1.json

{"nodeMap":{"q0":{"id":"q0","x":150,"y":240},"q1":{"id":"q1","x":430,"y":240},"q_accept":{"id":"q_accept","x":430,"y":60},"q_reject":{"id":"q_reject","x":700,"y":240}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0","targetid":"q0"},{"label":"1/1R,B/BR","sourceid":"q0","targetid":"q1"}],"q1":[{"label":"0/0R","sourceid":"q1","targetid":"q1"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","sourceid":"q1","targetid":"q_reject"}]},"finalNodes":["q_accept"],"nodeNum":4,"automataType":"TM","inputString":"1","stageProps":{"x":20,"y":60,"scale":1.6,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q6_broken_111.json

{"nodeMap":{"q0":{"id":"q0","x":150,"y":240},"q1":{"id":"q1","x":430,"y":240},"q_accept":{"id":"q_accept","x":430,"y":60},"q_reject":{"id":"q_reject","x":700,"y":240}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0","targetid":"q0"},{"label":"1/1R,B/BR","sourceid":"q0","targetid":"q1"}],"q1":[{"label":"0/0R","sourceid":"q1","targetid":"q1"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","sourceid":"q1","targetid":"q_reject"}]},"finalNodes":["q_accept"],"nodeNum":4,"automataType":"TM","inputString":"111","stageProps":{"x":20,"y":60,"scale":1.6,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q6_fixed_empty.json

{"nodeMap":{"q0":{"id":"q0","x":150,"y":240},"q1":{"id":"q1","x":430,"y":240},"q_accept":{"id":"q_accept","x":430,"y":60
},"q_reject":{"id":"q_reject","x":700,"y":240}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0","targetid":"q0"},
{"label":"1/1R","sourceid":"q0","targetid":"q1"},{"label":"B/BR","sourceid":"q0","targetid":"q_accept"}],"q1":[{"label":
"0/0R","sourceid":"q1","targetid":"q1"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","sourceid":"q1"
,"targetid":"q_reject"}]},"finalNodes":["q_accept"],"nodeNum":4,"automataType":"TM","inputString":"","stageProps":{"x":2
0,"y":60,"scale":1.6,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q6_fixed_11.json

{"nodeMap":{"q0":{"id":"q0","x":150,"y":240},"q1":{"id":"q1","x":430,"y":240},"q_accept":{"id":"q_accept","x":430,"y":60
},"q_reject":{"id":"q_reject","x":700,"y":240}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0","targetid":"q0"},
{"label":"1/1R","sourceid":"q0","targetid":"q1"},{"label":"B/BR","sourceid":"q0","targetid":"q_accept"}],"q1":[{"label":
"0/0R","sourceid":"q1","targetid":"q1"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","sourceid":"q1"
,"targetid":"q_reject"}]},"finalNodes":["q_accept"],"nodeNum":4,"automataType":"TM","inputString":"11","stageProps":{"x"
:20,"y":60,"scale":1.6,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q6_fixed_1.json

{"nodeMap":{"q0":{"id":"q0","x":150,"y":240},"q1":{"id":"q1","x":430,"y":240},"q_accept":{"id":"q_accept","x":430,"y":60
},"q_reject":{"id":"q_reject","x":700,"y":240}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0","targetid":"q0"},
{"label":"1/1R","sourceid":"q0","targetid":"q1"},{"label":"B/BR","sourceid":"q0","targetid":"q_accept"}],"q1":[{"label":
"0/0R","sourceid":"q1","targetid":"q1"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","sourceid":"q1"
,"targetid":"q_reject"}]},"finalNodes":["q_accept"],"nodeNum":4,"automataType":"TM","inputString":"1","stageProps":{"x":
20,"y":60,"scale":1.6,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}

q6_fixed_111.json

{"nodeMap":{"q0":{"id":"q0","x":150,"y":240},"q1":{"id":"q1","x":430,"y":240},"q_accept":{"id":"q_accept","x":430,"y":60
},"q_reject":{"id":"q_reject","x":700,"y":240}},"transitionMap":{"q0":[{"label":"0/0R","sourceid":"q0","targetid":"q0"},
{"label":"1/1R","sourceid":"q0","targetid":"q1"},{"label":"B/BR","sourceid":"q0","targetid":"q_accept"}],"q1":[{"label":
"0/0R","sourceid":"q1","targetid":"q1"},{"label":"1/1R","sourceid":"q1","targetid":"q0"},{"label":"B/BR","sourceid":"q1"
,"targetid":"q_reject"}]},"finalNodes":["q_accept"],"nodeNum":4,"automataType":"TM","inputString":"111","stageProps":{"x
":20,"y":60,"scale":1.6,"draggable":true},"stateOutput":{},"tapeMode":"single","inputString2":""}